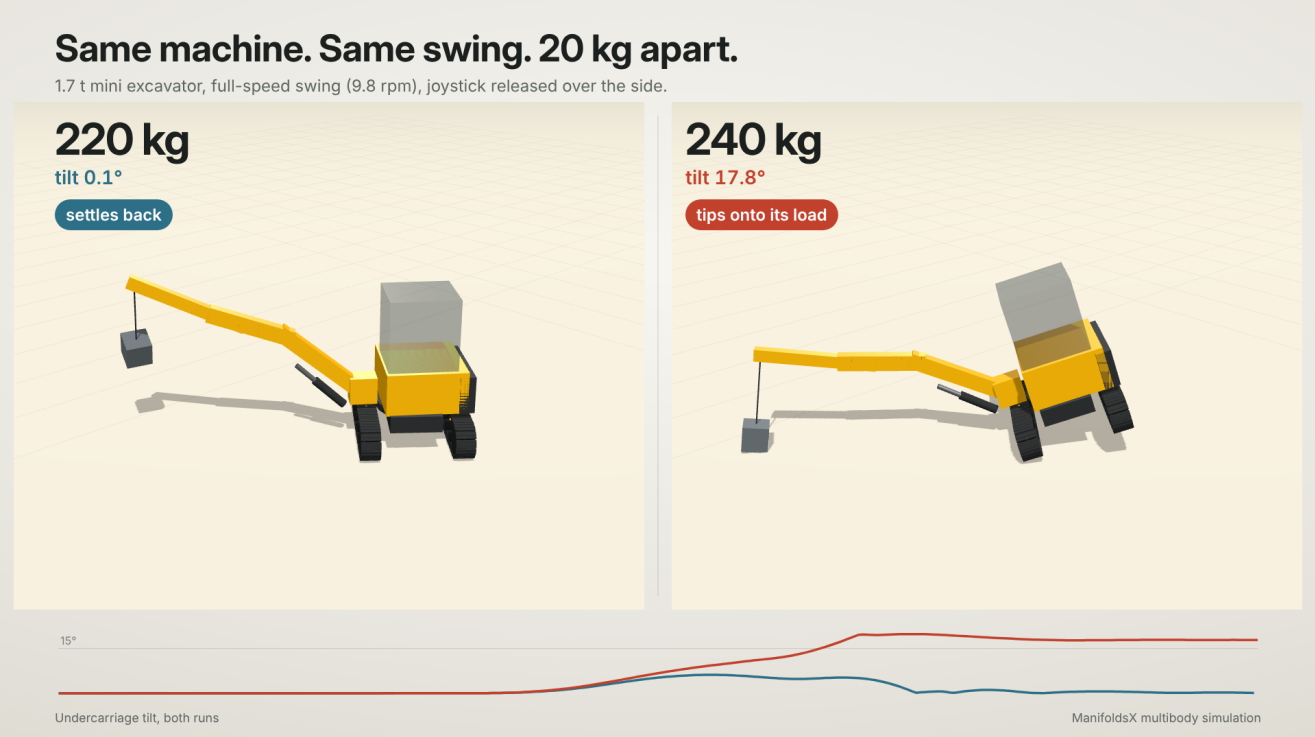
MANIFOLDSX / CONSTRUCTION MACHINERY

Mini excavator lift capacity, computed

A lift chart from public data, checked against the published one, and the cases a chart cannot rate

We built a multibody model of a 1.7 t mini excavator from the public specification sheet of the Cat 301.7 CR and computed its lift chart. Over the front, where the rating depends only on mass and geometry, the computed chart lands within about 10% of the published one. The same model then rates what a chart cannot: a slope, a load swinging on a sling when the upper stops, and how much each input matters.



Full-speed swing and stop at the lift point 2.0 m high, 3.13 m out. Left: 220 kg settles back. Right: 240 kg tips the machine onto its load.

Question	Result
Does the computed chart match the published one?	Over front, blade up: mean −6.7% (−12 to +2%). Over side: +11.2%. Blade down: −24.8% (limited by assumed cylinder geometry).
What do a slope and a swing stop do to capacity?	At 2.0 m and 3.13 m over the side: level 374 kg; 10° side slope 260 kg; full-speed swing and stop 229 to 278 kg.
Which inputs matter most?	Track gauge (±5% gives about ±6%), swing speed, then the upper's mass and centre of gravity. Boom and stick mass barely matter.
Are the numbers converged?	All lift and swing results agree with a fixed 1 ms reference within 0.2% over 30 simulated seconds.

The machine

The model is the whole machine. Each track is a loop of 38 shoes joined by revolute joints; each shoe meets the ground as a rounded box, and the road and carrier rollers ride on capsule rails along the shoes. The sprocket drives by friction and wrap, and a sprung idler tensions the track. On top sit the slewing upper, the boom, the stick and the bucket linkage, each cylinder modelled as a barrel and rod whose length is driven. For the lift and swing cases the bucket is removed, the cylinders are held at fixed lengths, and the load acts at the bucket pin, as on a lift chart.

Inputs from the public specification sheet

Quantity	Value
Minimum operating weight	1790 kg (model: 1796 kg)
Swing bearing height; tail swing radius	0.442 m; 0.65 m
Stick length, standard / long	0.96 m / 1.16 m
Track: shoe width, gauge, length	0.23 m on 1.07 m gauge (1.30 m expanded width); 1.59 m
Implement / swing relief pressure	245 bar / 147 bar
Stick digging force	9.5 kN (sets the stick cylinder bore, 50 mm)
Maximum lift radius at each chart height	reproduced by the boom foot (0.70, 0.69) m and a 1.80 m boom

Estimated or fitted

Quantity	Value	Basis
Upper, with operator and counterweight	840 kg; CG 0.35 m above the bearing	estimate
Upper CG, fore-aft	−0.006 m from the swing axis	fitted to one cell (std, 0 m, max radius, front)
Boom / stick	75 / 40 kg	estimate
Undercarriage frame / dozer blade	400 kg / 60 kg at 1.05 m	estimate
Boom cylinder bore	46.5 mm	fitted to the one visibly hydraulic-limited cell
Swing relief torque	3 and 5.5 kN·m	bracket from comparable machines

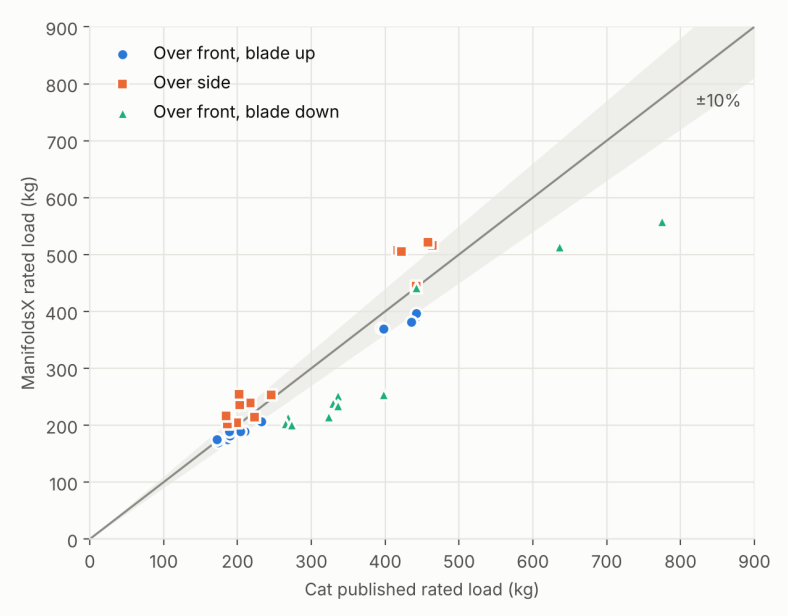
Contacts and solver

Ground contact is a Hunt–Crossley law (stiffness 2×10^{7}, exponent 1.5) with regularized Coulomb friction (ground 0.7, sprocket 0.6, regularization speed 5 mm/s). The equations are solved as an index-3 DAE with a generalized-α (HHT) integrator, Newton iteration with line search, a coupled contact tangent and a symmetric quasi-definite factorization of the KKT system. The step is error-controlled (tolerance 10^{-4}, capped at 4 ms); turning runs use 10^{-5} (section 5).

02 / VALIDATION

The published lift chart

For each of the 39 cells of the minimum-configuration chart (standard and long stick; over the front with the blade up and down; over the side), the bucket pin is placed at the cell, the cylinders are held, and a vertical load at the pin is ramped up slowly (40 kg/s, eight times slower once the machine has tilted 1.5°). The tipping load is the load at which the undercarriage has tilted 4° toward it: past the balance point the tilt grows quickly, and the slow ramp keeps the overshoot small. The hydraulic capacity is the load at which a cylinder reaches the 245 bar relief (full bore when pushing, annulus when pulling). The rated load is the smaller of 75% of tipping and 87% of hydraulic capacity, as manufacturer charts state.



Computed against published rated loads, 39 cells. The band is ±10%.

Column	Limit in the model	Computed vs published
Over front, blade up	tipping	mean −6.7%, range −12 to +2% (13 cells)
Over side	tipping or hydraulic	mean +11.2%, range −4 to +26% (13 cells)
Over front, blade down	hydraulic	mean −24.8%, range −36 to 0% (13 cells)

What the comparison says

- Over the front with the blade up, the rating depends only on masses and geometry. Public data and one fitted value reproduce it within about 10%.
- Over the side, the model tips at the outer track edge and reads up to 26% higher where tipping governs. A tipping line nearer the track centre would explain it; we did not check this against ISO 10567.
- With the blade down, the model is limited by our assumed cylinder geometry, and its shape follows that assumption. The real cylinder geometry would close this gap.
- The simulated tipping loads agree with a rigid-body moment balance of the same model within −7 to +2%. The simulation reads lower at short radius because the track system tilts about 2° before the balance point, which moves the load and the machine's centre of gravity toward the tipping edge. Appendix A lists every cell.

03 / BEYOND THE CHART

Slopes and swinging loads

A chart rates level, firm ground and a load held still. We took one published cell, the standard stick with the pin 2.0 m high at its maximum radius of 3.13 m, and ran the cases it cannot rate.

Slope

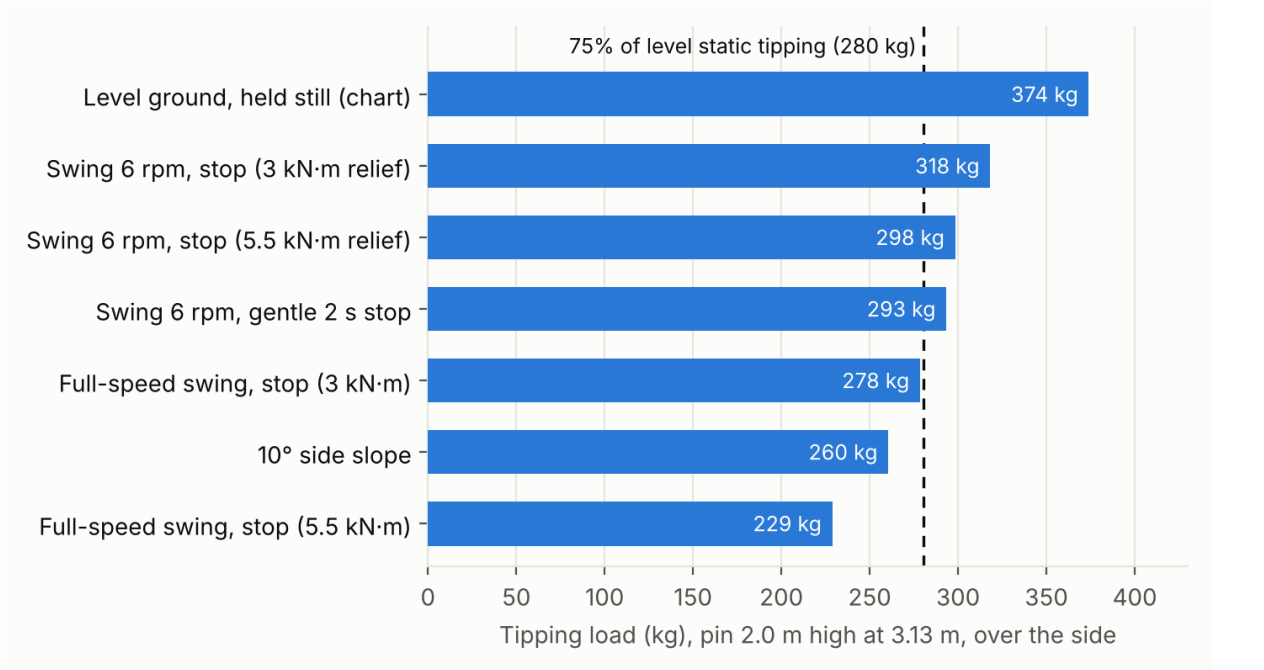
The ground stays horizontal and gravity is tilted by 10°, either across the machine (load downhill, over the side) or along it (facing downhill, over the front). The load acts along gravity. The same tipping search as for the chart is used.

Swing stop with a hanging load

The load hangs 1.0 m below the bucket pin on a spherical joint, so it swings freely and clears the ground. The machine settles with the boom over the front and the blade down, then swings 90° to the side: a smooth 1 s run-up to the commanded speed, a steady swing, and a stop commanded either over 2 s (gentle) or 0.2 s (joystick released).

How hard the upper can brake is set by the hydraulics, not by the command. The upper follows the commanded swing through a stiff spring and damper, standing for the oil in the swing motor lines (5 Hz on the swing inertia, 50% damping), and the spring slips whenever the torque would exceed the swing relief. Below the relief this reproduces a prescribed swing (peak tilt 0.710° against 0.708°, load swing 36.7° against 36.6°). The 301.7 swing relief is 147 bar, but its motor and gear are not public, so we bracket the torque: 3 kN·m, scaled from a 2.7 t machine that publishes 5.46 kN·m, and 5.5 kN·m, listed for a 1.85 t machine.

A run overturns when the undercarriage tilts 15° within 30 simulated seconds; at that tilt the load is about to touch the ground. The largest load that survives is found by bisection to 5 kg. Every limit was then run 5% heavier, and every one of those runs rolled past 40°, so the line is a point of no return. The load keeps swinging after the stop and rocks the machine; the largest tilt came as late as 7.4 s after the swing ended, which is why every run lasts 30 s.

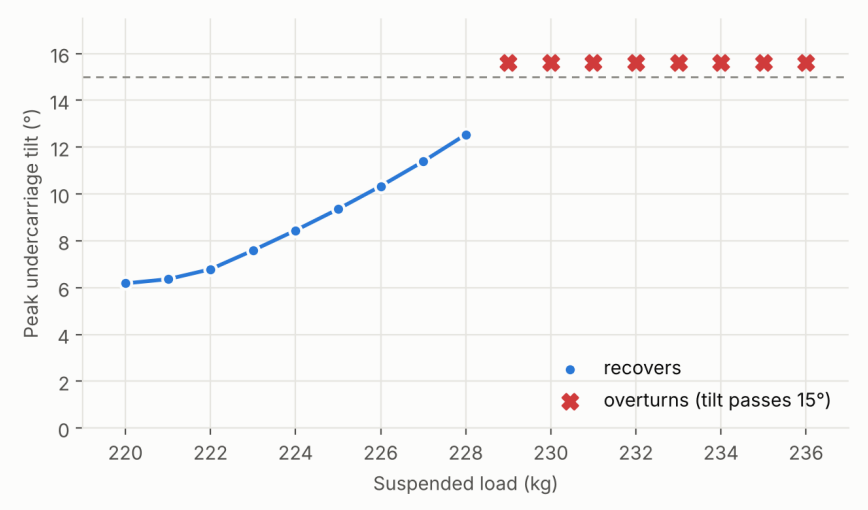


Tipping load over the side at 2.0 m and 3.13 m. The dashed line is 75% of the level static tipping load.

Case	Tipping load	vs level, side
Level, over side (chart condition)	374 kg	100%
Level, over front, blade up	275 kg	74%
Level, over front, blade down	754 kg	201%
10° side slope, load downhill	260 kg	70%
10° slope facing downhill, over front	184 kg	49%
Swing 6 rpm, gentle 2 s stop, 5.5 kN·m relief	293 kg	78%
Swing 6 rpm, joystick released, 3 kN·m	318 kg	85%
Swing 6 rpm, joystick released, 5.5 kN·m	298 kg	80%
Swing 9.8 rpm, joystick released, 3 kN·m	278 kg	74%
Swing 9.8 rpm, joystick released, 5.5 kN·m	229 kg	61%

Findings

- A rating at 75% of static tipping leaves a 25% margin. A 10° side slope uses 30% of it and a full-speed swing with a hard stop uses 26 to 39%, depending on the relief torque: either alone uses up the margin. A swing at 6 rpm uses 15 to 22%.
- The swing torque decides the full-speed result: 229 kg with a 5.5 kN·m relief, 278 kg with 3 kN·m.
- The start of the swing sets the load swinging. A lower relief starts the swing more gently (load swing 24° instead of 36° at 6 rpm), which is why a gentle 2 s stop at full relief (293 kg) holds less than releasing the joystick with a 3 kN·m relief (318 kg).
- Facing downhill on a 10° slope, the machine tips over the front at 67% of its level-ground load.



Peak tilt against load for the full-speed swing and stop, 5.5 kN·m relief, 1 kg steps.

Near the limit the outcome is a cliff. The peak tilt rises smoothly from 6.2° at 220 kg to 12.5° at 228 kg; from 229 kg on, every run overturns. So bisection on the limit is well posed, and the limit is sharp to about 1 kg for this model.

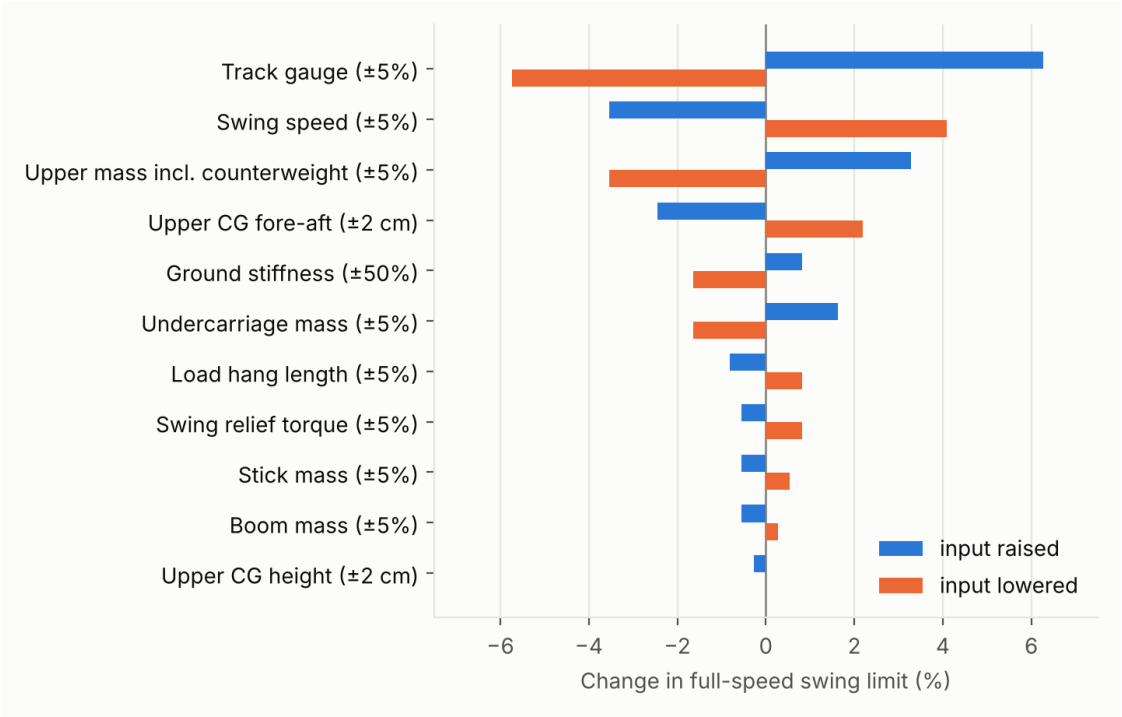
What happens after 15°

The rating runs have no contact between the load and the ground. With that contact switched on, the 240 kg run tilts to 20°, the load lands, and the machine stays propped at about 18° on its load with the near track in the air, as in the clip on page 1. The overturn line at 15° sits just before the load would touch down, so the ratings do not depend on it.

04 / SENSITIVITY

Which inputs matter

At the same cell we changed one input at a time, down and up, and recomputed the static tipping loads over the side (374.0 kg) and the front (274.9 kg) and the full-speed swing limit with the 5.5 kN·m relief (229.1 kg, bisection to 1 kg). Masses, lengths, torque and speed were nudged by ±5%, centre-of-gravity positions by ±2 cm and the ground stiffness by ±50%: 218 simulations.



Change in the full-speed swing limit for each input nudged down and up.

Input	Nudge	Side rating	Front rating	Swing limit
Track gauge	±5%	−6.1 / +6.3%	+0.0 / +0.0%	−5.7 / +6.3%
Swing speed	±5%	–	–	+4.1 / −3.5%
Upper mass incl. counterweight	±5%	−2.8 / +2.8%	−3.2 / +3.2%	−3.5 / +3.3%
Upper CG fore-aft	±2 cm	+1.8 / −1.8%	+2.4 / −2.4%	+2.2 / −2.5%
Undercarriage frame mass	±5%	−1.3 / +1.3%	−1.6 / +1.6%	−1.6 / +1.6%
Ground stiffness	±50%	−1.8 / +0.6%	−1.3 / +0.4%	−1.6 / +0.8%
Swing relief torque	±5%	–	–	+0.8 / −0.5%
Load hang length	±5%	–	–	+0.8 / −0.8%
Boom mass	±5%	+0.3 / −0.3%	+0.5 / −0.5%	+0.3 / −0.5%
Stick mass	±5%	+0.4 / −0.4%	+0.6 / −0.5%	+0.5 / −0.5%
Upper CG height	±2 cm	+0.0 / −0.0%	+0.1 / −0.1%	+0.0 / −0.3%

Track gauge dominates the side and swing ratings: each 1% of gauge is worth about 1.2% of capacity. The 301.7 undercarriage is expandable, and this cell is at the expanded gauge. Swing speed is the strongest lever an operator has: swinging 5% slower raises the limit by 4%. The upper's mass and the fore-aft position of its centre of gravity come next, and those are the machine data a model most needs. Boom and stick mass, the height of the upper's centre of gravity and the load's hang length each move the result by less than 1%. The relief torque is flat locally (±5% moves the limit by under 1%) but not over its range: 3 against 5.5 kN·m is 278 against 229 kg.

What changes with the time step?

Each configuration was run for 30 simulated seconds twice with identical inputs: at the default error-controlled step (tolerance 10^{-4}, capped at 4 ms), which every result above uses, and at a fixed 1 ms step.

Run	Output	Default	1 ms	Difference
Straight drive, 0.5 m/s	mean speed, 10–30 s	0.4986 m/s	0.4993 m/s	−0.1%
12° ramp	height climbed	3.091 m	3.091 m	0.0%
Dig cycle	peak boom / stick / bucket cylinder force	28.80 / 15.82 / 10.09 kN	28.81 / 15.78 / 10.09 kN	≤ 0.2%
Lift, over side, std 2.0 m max	tipping load	374.0 kg	374.0 kg	0.0%
Swing stop 220 kg	peak tilt	6.182°	6.175°	+0.1%
Swing stop limit	overturns at	229 kg	225 kg recovers, 233 kg overturns	brackets
Skid-steer turn	heading change in 30 s	189.7°	159.3°	+19%

Constraint violation stays below 3×10^{-12} in every run.

Turning

The turn is the exception. In a skid-steer turn the shoes slide sideways, and those near the turn centre slide slowly, inside the friction regularization. The steady turn rate converges at first order in the step, and the error estimate, which is position based, does not see the bias, so the controller keeps 2–4 ms steps:

Step	Turn rate, 8–14 s	vs converged
default (tolerance 10^{-4}, cap 4 ms)	7.30°/s	+27%
fixed 2 ms	7.10°/s	+24%
fixed 1 ms	6.09°/s	+6%
tolerance 10^{-5}	5.96°/s	+4%
fixed 0.5 ms	5.91°/s	+3%
fixed 0.25 ms	5.82°/s	+2%
extrapolated (first order)	5.73°/s	

Turning runs now use a tolerance of 10^{-5}, at about 1.35 times the cost. A velocity-level term in the error estimate is the planned fix. The lift and swing results do not depend on this: their tracks do not slide before the machine tips, and they agree with the 1 ms reference above.

06 / LIMITS

What this study does not show

- Masses and centres of gravity are estimates. One value, the fore-aft position of the upper's centre of gravity, is fitted to one published cell.
- The swing motor torque and the cylinder geometry are not public. We bracket the torque and assume the geometry, which is why the blade-down column is off by about 25%.
- The ground is firm: a contact stiffness and friction, no sinkage. The operator makes no corrections during the swing.
- The dynamic cases and the sensitivity study use one chart cell. Other cells, the long stick, other slopes and a retracted undercarriage were not run.
- Overturn in the dynamic cases means a 15° tilt within 30 s; the static chart uses the balance point (4° tilt on a slow ramp). The two are different events and are not compared directly.

What would close the gaps

Ranked by the sensitivity study: the upper's mass and centre of gravity; the swing motor displacement and gear ratio (the swing torque); the cylinder pin positions and bores. With those, the same process gives a full computed chart for every configuration, and the stability limits beyond it, for a fraction of the cost of testing each variant.

Independence

This is independent work by ManifoldsX, not affiliated with or endorsed by Caterpillar. Cat and 301.7 identify the public specification sheet and lift chart used as reference data. Scripts and result files are available on request: manifoldsx.eng@gmail.com.

APPENDIX A

The chart, cell by cell

Rated = min(75% of tipping, 87% of hydraulic capacity). Rigid = moment balance of the same model about the outermost loaded contact. Loads in kg; h and r are the bucket pin height and radius. Bores: stick 50 mm, boom 46.5 mm; relief 245 bar.

Stick	h (m)	r (m)	Column	Tipping	Rigid	Hydraulic	Rated	Published	Diff.	Limit
long	2.0	3.35	front, blade down	680	679	246	214	268	−20%	hydraulic
long	2.0	3.35	front, blade up	252	262	246	189	210	−10%	tipping
long	2.0	3.35	side	342	340	246	214	223	−4%	hydraulic
long	1.5	3.54	front, blade down	626	621	234	204	265	−23%	hydraulic
long	1.5	3.54	front, blade up	234	240	234	175	186	−6%	tipping
long	1.5	3.54	side	318	313	235	204	199	+3%	hydraulic
long	1.0	2.00	front, blade down	1668	1704	591	514	636	−19%	hydraulic
long	1.0	2.00	front, blade up	520	551	590	390	441	−12%	tipping
long	1.0	2.00	side	707	722	593	516	464	+11%	hydraulic
long	1.0	3.64	front, blade down	601	595	231	201	273	−26%	hydraulic
long	1.0	3.64	front, blade up	226	230	232	169	174	−3%	tipping
long	1.0	3.64	side	306	301	232	202	186	+9%	hydraulic
long	0.0	2.00	front, blade down	1629	1639	790	688	946	−27%	hydraulic
long	0.0	2.00	front, blade up	494	510	799	371	394	−6%	tipping
long	0.0	2.00	side	676	678	798	507	417	+22%	tipping
long	0.0	3.58	front, blade down	619	610	247	215	323	−33%	hydraulic
long	0.0	3.58	front, blade up	234	236	248	175	172	+2%	tipping
long	0.0	3.58	side	315	308	248	216	185	+17%	hydraulic
std	2.0	3.13	front, blade down	754	756	290	252	336	−25%	hydraulic
std	2.0	3.13	front, blade up	275	287	290	206	232	−11%	tipping
std	2.0	3.13	side	374	373	291	253	246	+3%	hydraulic
std	1.5	2.00	front, blade down	1683	1731	508	442	442	+0%	hydraulic
std	1.5	2.00	front, blade up	529	568	506	397	442	−10%	tipping
std	1.5	2.00	side	719	739	510	444	442	+0%	hydraulic
std	1.5	3.33	front, blade down	682	680	274	239	329	−27%	hydraulic
std	1.5	3.33	front, blade up	252	260	275	189	204	−7%	tipping
std	1.5	3.33	side	342	339	275	239	217	+10%	hydraulic
std	1.0	2.00	front, blade down	1650	1686	642	558	775	−28%	hydraulic
std	1.0	2.00	front, blade up	508	539	642	381	435	−12%	tipping
std	1.0	2.00	side	695	709	645	521	458	+14%	tipping
std	1.0	3.44	front, blade down	651	646	271	235	336	−30%	hydraulic
std	1.0	3.44	front, blade up	242	248	271	181	190	−5%	tipping
std	1.0	3.44	side	328	323	271	236	203	+16%	hydraulic
std	0.0	2.00	front, blade down	1625	1635	789	687	938	−27%	hydraulic
std	0.0	2.00	front, blade up	492	508	799	369	398	−7%	tipping
std	0.0	2.00	side	674	675	797	505	422	+20%	tipping
std	0.0	3.37	front, blade down	673	666	292	254	398	−36%	hydraulic
std	0.0	3.37	front, blade up	252	255	293	189	189	−0%	tipping
std	0.0	3.37	side	339	332	293	254	202	+26%	tipping